EVALUATION REPORT / SUPERVISED RECORDINGS

Multi-Session Transfer on C-CO Recordings

Source pretraining: C-CO0 • Target adaptation: C-CO1 to C-CO52 • Train/holdout: approximately 80/20 • Seeds: 42 and 43

01	52	02	832
SOURCE / C-CO0	TARGET SESSIONS	RANDOM SEEDS	TARGET EVALUATIONS

Protocol

A shared CEBRA/ACORN backbone was trained on the training partition of C-CO0. For each target session, the backbone was kept fixed while session-specific input/output Conv1D adapters and a GRU behavioral decoder were fitted on that session's training data. Performance was evaluated as clean behavioral-decoding R^2 on the provided held-out partition. The nominal split is approximately 80% training / 20% holdout; the temporal ordering of the original partitions has not been independently confirmed.

Training configuration: 3,000 source-encoder steps; 1,000 target-adapter steps; 2,500 decoder steps; batch size 512. All eight conditions below contain 52 targets × 2 seeds = 104 target evaluations, with successful results for every condition.

Overall performance

Training condition	Source R^2	Target mean R^2	Target median R^2	Target R^2 < 0.05
Noise-shaped GAUSS	**0.8014**	**0.7996**	**0.8150**	**0/104**
CEBRA (standard)	0.8076	0.7719	0.7937	0/104
Adaptive d′-PGD (behavior)	**0.7000**	**0.6644**	**0.6769**	**0/104**
Adaptive d′-PGD (signal)	0.7066	0.6577	0.6760	0/104
Fixed d′-PGD (ε=8)	0.7083	0.6528	0.6803	0/104
Adaptive pooled covariance	0.7308	0.6277	0.6511	0/104
L2-PGD (ε=3)	0.7771	0.2167	-0.0006	66/104
L2-PGD (ε=6)	0.8071	0.1298	-0.0012	82/104

Top-line findings

01 Noise-shaped GAUSS achieved the highest clean transfer score (mean R^2 0.7996), exceeding standard CEBRA by +0.0277 R^2.

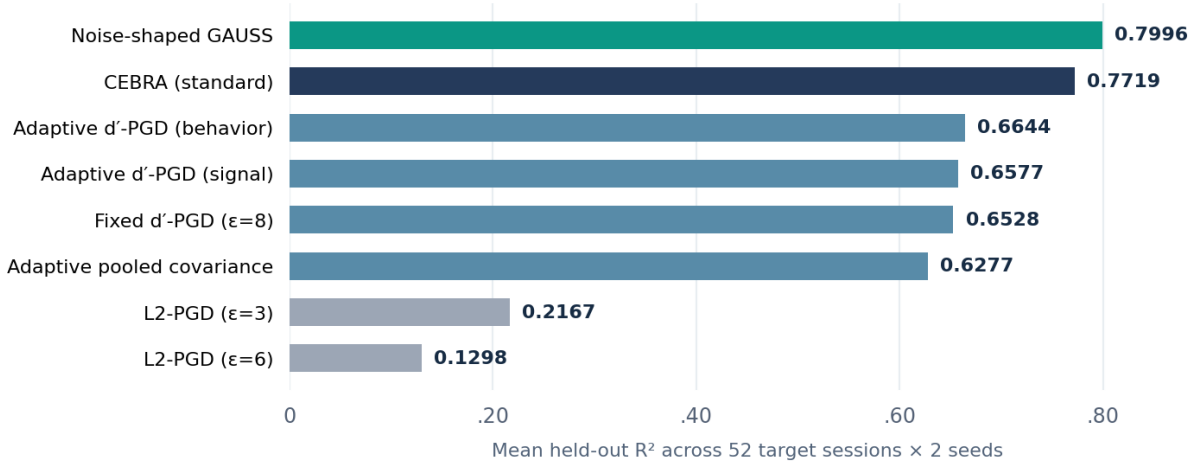
02 Higher-budget L2-PGD showed frequent near-zero held-out R^2 (66/104 and 82/104 for ε=3 and 6); the noise-aware conditions had no cases below 0.05.

03 Adaptive behavior-based d′-PGD reached R^2 0.6644 versus 0.6528 for fixed d′=8, a modest +0.0116 difference.

QUANTITATIVE COMPARISON / HELD-OUT DECODING

Matched Comparisons and Failure Analysis

All differences below are computed between the same target session and seed, so method comparisons are paired rather than based on unmatched subsets.



Paired differences in target-session R^2

Matched comparison	Mean ΔR^2	Wins / 104	Session wins / 52
Noise-shaped GAUSS vs CEBRA	**+0.0277**	**90/104**	**48/52**
Adaptive behavior vs Fixed d′=8	**+0.0116**	**55/104**	**27/52**
Adaptive signal vs Fixed d′=8	+0.0049	48/104	26/52
Fixed d′=8 vs L2-PGD ε=3	+0.4361	82/104	46/52
Fixed d′=8 vs L2-PGD ε=6	+0.5230	89/104	47/52
Adaptive behavior vs L2-PGD ε=6	+0.5346	91/104	48/52

Session-level bootstrap (20,000 resamples; both seeds averaged within each session): GAUSS − CEBRA = +0.0277 R^2, 95% interval [+0.0130, +0.0441]; adaptive behavior − fixed d′=8 = +0.0116, 95% interval [−0.0033, +0.0270]. The latter interval overlaps zero.

Near-zero performance diagnostic

Training condition	Train $R^2 < 0.05$	Holdout $R^2 < 0.05$	Holdout $R^2 < 0$
L2-PGD (ε=3)	59 / 104	66 / 104	59 / 104
L2-PGD (ε=6)	80 / 104	82 / 104	73 / 104
Noise-aware PGD family (4 conditions)	**0 / 416**	**0 / 416**	**0 / 416**
Noise-shaped GAUSS	0 / 104	0 / 104	0 / 104
Standard CEBRA	0 / 104	0 / 104	0 / 104

BUDGET CALIBRATION / SESSION VARIABILITY

Noise Budgets and Session-Level Outcomes

Training budgets for covariance-aware PGD

Condition	Min ε	Median ε	Max ε
Fixed d′-PGD (ε=8)	8.000	8.000	8.000
Adaptive d′-PGD (behavior)	**6.253**	**8.141**	**12.626**
Adaptive d′-PGD (signal)	5.725	7.513	12.626
Adaptive pooled covariance	3.760	6.814	10.388

The fixed covariance-aware condition uses d′=8 throughout. Adaptive settings vary their budgets by session; behavior-based calibration spans 6.253–12.626. GAUSS uses random covariance-shaped perturbations rather than adversarial PGD; its logged noise scale (median 46.861) is not directly comparable to the d′-PGD budgets.

Illustrative session results

Target	CEBRA	Fixed d′=8	Auto behavior	Auto signal	GAUSS
C-CO1	0.5873	0.3832	0.4499	0.4658	0.6557
C-CO5	0.7701	0.6469	0.6119	0.6064	0.7938
C-CO23	0.4980	0.4675	0.4607	0.4459	0.5009
C-CO25	0.8325	0.7309	0.7995	0.7910	0.8472
C-CO39	0.7678	0.6219	0.5705	0.5690	0.7841
C-CO45	**0.4484**	**0.5970**	**0.6421**	**0.5595**	**0.7680**
C-CO52	0.8827	0.8216	0.8018	0.6832	0.8866

Values above are averages over the two seeds. C-CO45 illustrates a large gap between standard CEBRA (0.4484) and GAUSS (0.7680), whereas in C-CO52 both are high (0.8827 and 0.8866). Improvements are not uniform across sessions.

Interpretation and scope

RESULT Clean transfer: GAUSS is best overall. Covariance-aware adversarial training does not exceed standard CEBRA in mean held-out R^2.

RESULT Training stability: covariance-aware PGD avoids the frequent near-zero R^2 seen with larger L2-PGD budgets, although the attack geometries are not matched by perturbation strength.

RESULT Adaptive ε: behavior-based adaptation slightly improves average R^2 over the fixed d′ budget, but the uncertainty interval does not establish a robust advantage.

Scope limits: The target sessions use fitted adapters and decoders, so this is calibrated transfer rather than zero-shot generalization. Results are clean held-out R^2, not target-session adversarial robustness. The supplied train/holdout partitions are nominally ~80/20, but their chronological ordering is unverified. Only two seeds and the C-CO recording family are covered.